Level 2

Hackathon – Fundamentals of Web Development

Student Details

Student Name	Murshida Fathima
Register Number	34625U23013
Department	B.Sc. Information System Management
ID	5C6BDA10F6FB641BDBC469CEBFA023C8
College	M.M.E.S. Women's Arts and Science College
Date of Submission	27.10.2026

Git-Hub Link:

1. Problem Statement

Traditional quizzes are usually conducted using paper, which takes more time to prepare, conduct, and evaluate. Students may also have to wait for their results. The Online Quiz Application provides a simple digital platform where users can answer multiple-choice questions and receive their scores instantly. It makes quiz participation easier, faster, and more interactive.

2. Objectives

1. To develop an easy-to-use online quiz application.
2. To display multiple-choice questions to users.
3. To allow users to select answers.
4. To calculate scores automatically.
5. To display the final result immediately.
6. To improve the learning and assessment experience.

3. Architecture

The Online Quiz Application uses HTML, CSS, and JavaScript to create an interactive website.

HTML creates the structure of the application. CSS improves its appearance, layout, and colours. JavaScript manages questions, answer selection, navigation, and score calculation.

Application Flow

Open Website
↓
Home Page
↓
Display Question
↓
Select Answer
↓
Click Next
↓
Check Answer
↓
Calculate Score
↓
Display Final Result
↓
Restart Quiz

Technologies Used

HTML: Creates the structure and content of the webpage.

CSS: Styles the webpage, buttons, backgrounds, and question area.

JavaScript: Displays questions, checks answers, and calculates the score. Web Browser: Runs and displays the application.

4. Practical Implementation

The Online Quiz Application was developed using HTML, CSS, and JavaScript. HTML provides the webpage structure, CSS makes the interface attractive, and JavaScript controls the quiz functionality.

Users can view questions, select answers, move to the next question, and see their final scores after completing the quiz.

4.1 Features Implemented

1. Home Page

Provides access to the online quiz and introduces the application.

2. Quiz Questions

Displays questions and multiple-choice answers to the user.

3. Answer Selection
Allows users to select an answer from the available options.
4. Next Button
Moves the user to the next question after an answer is selected.
5. Answer Checking
Checks whether the selected answer matches the correct answer.
6. Score Calculation
Automatically counts the number of correct answers.
7. Result Display
Displays the final score after the quiz is completed.







4.2 Source Code

HTML

HTML is used to create the structure of the Online Quiz Application. It contains the heading, question area, answer options, Next button, and result section.

Important HTML code:

```
<!DOCTYPE html>
<html lang=”en”>
<head>
  <meta charset=”UTF-8”>
  <meta name=”viewport”
        Content=”width=device-width, initial-scale=1.0”>
  <title>Online Quiz Application</title>
  <link rel=”stylesheet” href=”style.css”>
</head>
<body>
  <div class=”quiz-container”>
    <h1>Online Quiz</h1>
    <p id=”question-number”></p>
    <h2 id=”question”>Question</h2>
    <div id=”options”></div>
    <button id=”next-btn”>Next</button>
    <p id=”result”></p>
```

```
  </div>

  <script src=”script.js”></script>
</body>
</html>
```

CSS

CSS is used to improve the appearance of the application. It controls the background, font, spacing, buttons, and quiz container. Important CSS code:

- {

```
  Box-sizing: border-box;
}

Body {
  Margin: 0;
  Font-family: Arial, sans-serif;
  Background: linear-gradient(135deg, #e8ddff, #dce9ff);
  Min-height: 100vh;
  Display: flex;
  Justify-content: center;
  Align-items: center;
}

.quiz-container {
  Background: white;
  Padding: 25px;
  Border-radius: 15px;
  Width: 90%;
  Max-width: 500px;
  Text-align: center;
}

Button {
  Padding: 10px 20px;
  Border: none;
  Border-radius: 8px;
  Cursor: pointer;
}
```

JavaScript

JavaScript controls the main functions of the quiz. It displays questions, records selected answers, checks correctness, and calculates the final score.

Important JavaScript code:

```
let currentQuestion = 0;
let score = 0; let
selectedAnswer = null;
function showQuestion() {
selectedAnswer = null;

  questionNumber.innerHTML =
"Question " + (currentQuestion + 1) +
    " of " + questions.length;

  question.innerHTML =     questions[currentQuestion].question;

  options.innerHTML = "";

  questions[currentQuestion].options.forEach(
(option, index) => {       const button =
document.createElement("button");

     button.innerHTML = option;       button.classList.add("option");

     button.onclick = function () {
selectedAnswer = index;
     };

     options.appendChild(button);
   }
  );
}

nextButton.onclick = function () {   if
(selectedAnswer === null) {
alert("Please select an answer!");
return;
  }

  if (selectedAnswer ===
questions[currentQuestion].answer) {     score++;
  }
```

```
  currentQuestion++;

  if (currentQuestion < questions.length) {     showQuestion();
  } else {     questionNumber.innerHTML =
"";     question.innerHTML = "Quiz Completed!";
options.innerHTML = "";
nextButton.style.display = "none";

    result.innerHTML =       "Your
Score: " + score +
      " / " + questions.length;
  }
};

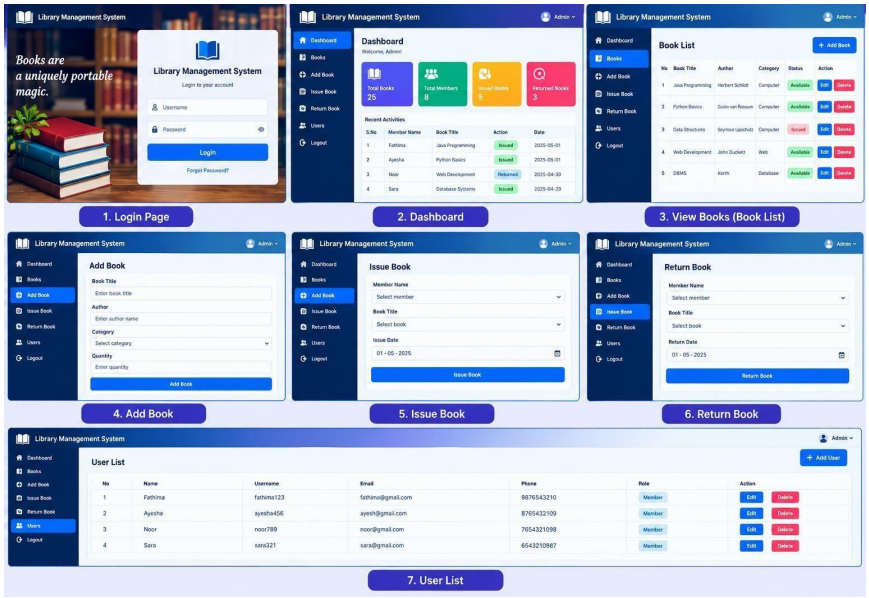
showQuestion();
```

Note: These are important code sections. The complete JavaScript program also needs the question data and the declarations that connect the HTML elements to JavaScript.

4.3 Project Execution

1. Save the HTML code in a file named `index.html`.
2. Save the CSS code in a file named `style.css`.
3. Save the JavaScript code in a file named `script.js`.
4. Keep all three files in the same folder.
5. Open `index.html` in a web browser.
6. Select an answer and click the Next button.
7. Continue answering until all questions are completed. View the final score displayed on the scr

5. Output Screenshots



6. Technical Explanation

HTML

HTML provides the basic structure of the application. It creates headings, question areas, answer options, buttons, and result sections.

CSS

CSS improves the visual appearance of the application. It controls colours, fonts, spacing, buttons, and the layout of the quiz interface.

JavaScript

JavaScript makes the application interactive. It displays questions, records selected answers, checks correctness, and calculates the final score.

Event Handling

Event handling allows the application to respond when users click answer options or the Next button.

Score Calculation

The application increases the score when the selected answer is correct. After all questions are answered, it displays the total score.

7. Conclusion

The Online Quiz Application provides a simple and interactive way to conduct quizzes digitally. Users can answer multiple-choice questions and view their scores immediately. This project helped develop practical knowledge of HTML, CSS, JavaScript, webpage design, and event handling.

8. Future Enhancements

1. Add user registration and login.
2. Introduce different quiz categories.
3. Add a timer for each quiz.

4. Include more questions and difficulty levels.
5. Store scores in a database.
6. Add a leaderboard to display top scores.
7. Provide explanations for correct answers.
8. Create an administrator panel to manage questions.
9. Improve the interface for mobile devices.